

Supervisor Meeting Notes — Week 6

Classification of Malignant Melanoma in Canines (CATCH) · DAIM2025A_088

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1. Done this fortnight

- Attention U-Net (Oktay 2018) + 3-way segmentation comparison.
- ResNet-50 tumour-subtype classifier (transfer learning).
- Pipeline made real-CATCH-ready: .svs reading, COCO/SQLite annotation masks, 7 subtypes, slide-level split.

2. Honest data status

No real CATCH slides processed yet — results are on **synthetic** data and validate the code, not the biology. Real set: 350 .svs WSIs, 50 each of 7 subtypes, 12,424 polygon annotations (TCIA DOI 10.7937/TCIA.2M93-FX66). Blocker: license-gated, ~hundreds-of-GB download; needs OpenSlide.

3. Three bugs found and fixed

- **Label bug**: subtype guessed from a filename colour-cast → now from the real subtype folder/annotation.
- **Data leakage**: patch split mixed same-slide patches across train/test → now slide-level (verified zero overlap).
- **Transfer learning**: frozen-encoder BatchNorm kept updating → now held in eval mode.

4. Results (synthetic — pipeline check, not biology)

Model	Params	Dice	IoU	Hausdorff
Baseline U-Net	31.04M	0.842	0.727	111.3
U-Net + ResNet-34	24.44M	0.225	0.127	218.5
Attention U-Net	31.39M	0.965	0.932	113.3

ResNet-50 (demo, 3 classes): accuracy=1.000, macro-F1=1.000, AUC=1.000 — perfect score is a synthetic-data artefact.

5. Decisions I need from you

- 7-class subtype vs. binary melanoma-vs-rest as primary target?
- Combined end-to-end metric vs. separate seg/cls evaluations?
- Data scope: a subset of subtypes to start, or wait for all 350?
- Single split now, 5-fold CV reserved for the final?

6. Next fortnight (W7-8)

- EfficientNet-B3 + comparison with ResNet-50.
- First real-data run once OpenSlide + download are in place.
- Evaluation phase: 5-fold CV, McNemar's test, Grad-CAM.